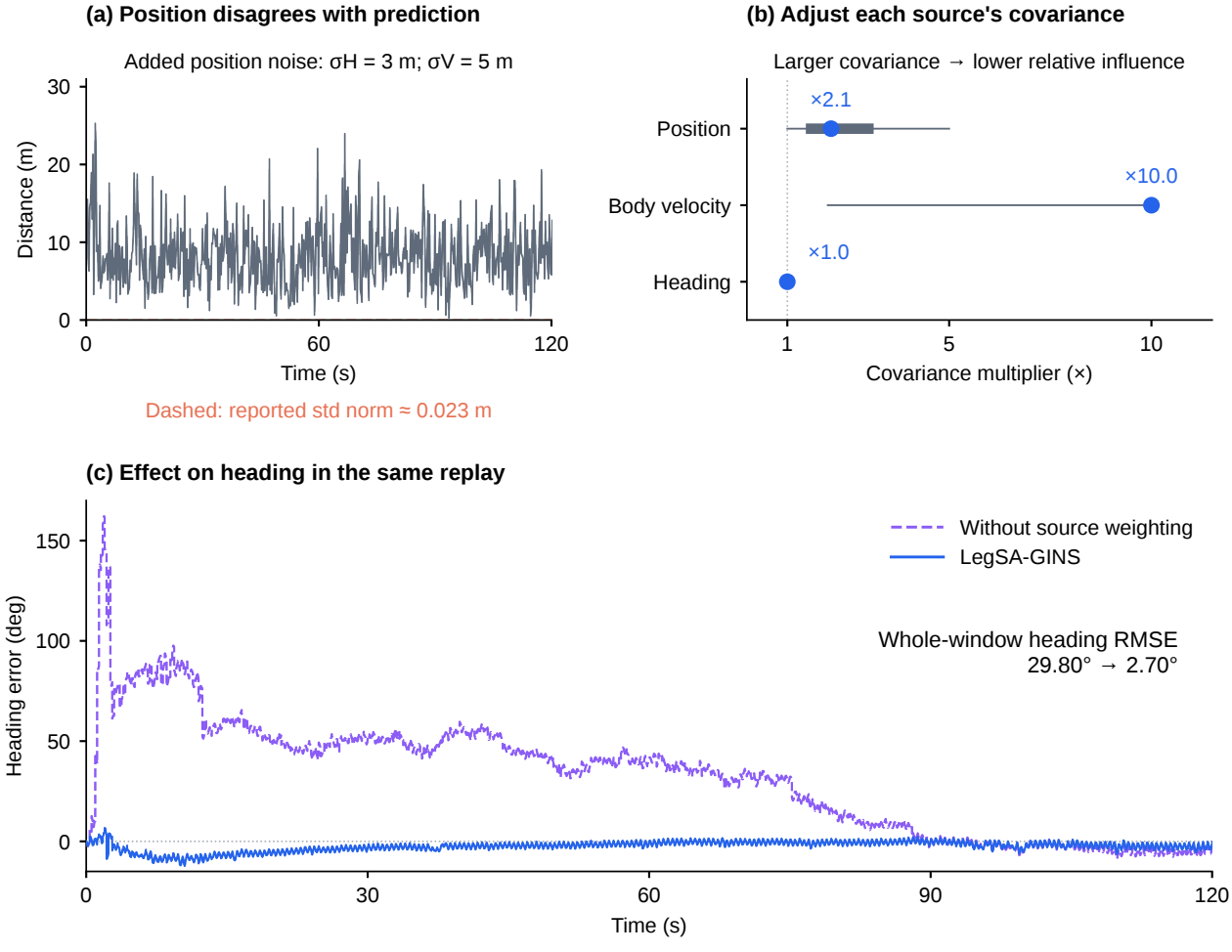

(a) Position disagrees with prediction
Added position noise: σH = 3 m; σV = 5 m
Distance (m)
30
20
10
0
0
60
120
Time (s)
Dashed: reported std norm ≈ 0.023 m
(b) Adjust each source's covariance
Larger covariance → lower relative influence
Position
Body velocity
Heading
×2.1
×10.0
×1.0
1
5
10
Covariance multiplier (×)
(c) Effect on heading in the same replay
Heading error (deg)
150
100
50
0
0
30
60
90
120
Time (s)
Without source weighting
LegSA-GINS
Whole-window heading RMSE
29.80° → 2.70°
Weighting off → full, over 274 s: horizontal RMSE 3.17 → 3.56 m.
The heading benefit in this replay carries a positioning cost.